

The Profile Bulging PyRoll Plugin

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This plugin predicts the free-surface ("bulge" / rounding) shape of a profile's cross-section after a rolling pass, i.e. the part of the boundary that is not in contact with the rolls. It implements analytical models from the literature for both two-roll and three-roll passes, with a particular focus on three-roll (Kocks mill / flat-roll block) passes, which is the subject of this restructured version of the plugin.

1 Model scope

All models implemented here predict the free-surface shape *given* the maximum width B_1 of the profile after rolling. How B_1 itself is predicted (spread prediction) is a separate concern, handled by other PyRoll plugins such as `pyroll-wusatowski-spreading`; this plugin only reshapes the free surface of the cross-section consistently with a given (already solved) width. This mirrors how the source papers themselves separate the two problems: a spread formula calibrated against FE data, and a free-surface-profile formula that consumes the predicted spread as an input.

A generic spread formula such as Wusatowski's needs the equivalent height and width of the incoming and outgoing profile, which `pyroll-core` does not define for arbitrary (in particular 3-fold symmetric) cross-sections on its own; `pyroll-lendl-equivalent-method` supplies them via Lendl's equivalent rectangle method, including a `ThreeRollPass`-specific formula. Loading it together with a spread-prediction plugin lets a `ThreeRollPass` predict its own out-profile width instead of requiring it to be set explicitly, and is recommended for realistic use; Section 4 demonstrates the full chain (Lendl + Wusatowski + this plugin) end to end.

2 Three-roll passes

Three-roll (Kocks) mills roll a workpiece between three rolls arranged 120° apart about the pass line, closing in on the stock symmetrically. Two variants are modeled, following Min, Kwon, Lee, Woo, and Im [1] and Byon, Kim, Kim, and Lee [2] respectively:

- **Flat-roll mills** [1]: the roll surface is flat along its axis (no machined groove); a round bar is progressively formed into a hexagonal shape over several passes.

pyroll-core's FlatGroove used in a ThreeRollPass models this case ("flat" classifier).

- **Kocks mills with a curved (grooved) roll surface** [2]: the roll surface carries a shallow, machined groove (radius $D_g/2$), producing a rounded ("Koval") caliber. Any non-flat groove used in a ThreeRollPass (round, oval, hexagonal, ...) is treated this way.

Both papers distinguish three types of geometry transition, based on the shape of the incoming profile and of the current pass:

1. **round** $\rightarrow$ **curved-hexagonal** / **Koval** – the incoming profile is round ("round" in its classifiers).
2. **curved-hexagonal** $\rightarrow$ **hexagonal** / **Koval** $\rightarrow$ **Koval** – neither the incoming profile nor the current pass is round.
3. **Koval** $\rightarrow$ **round** – the current pass itself is round (the final pass of a Kocks mill schedule). Min et al. do not treat this case, since their experiments never return to a round shape.

2.1 Flat-roll mills (Min et al. 2003)

For the round $\rightarrow$ curved-hexagonal pass, the free surface is modeled as a circular arc, whose center of curvature is offset from the profile's center by an eccentricity e :

$$\frac{e}{B_0} = 2.40 \cdot \frac{F_H}{F_O}, \quad R_1 = B_1 - e,$$

where B_0 is the incoming profile's own radius (half its maximum width), B_1 the radius of the outgoing profile after rolling, F_O the incoming profile's cross-sectional area and F_H the part of it that does not fit the new pass's usable cross-section (`roll_pass.displaced_cross_section` in `pyroll-core`), i.e. the area the roll must displace.

For the curved-hexagonal $\rightarrow$ hexagonal and hexagonal $\rightarrow$ hexagonal passes, Min et al. approximate the free surface as a *straight chord* instead, from a point W_1 on the roll face (where contact ends) to the corner tip B_1 . This plugin locates W_1 as the point where the roll face crosses the circle of radius B_1 around the profile's center, on each of the two roll faces flanking a given corner, and builds the polygon directly from the resulting W_1, B_1, W_1 vertex triples at each of the three corners; between corners the boundary follows the (straight) roll face exactly, requiring no further approximation there. Min et al. note this linear approximation slightly under-predicts the true (mildly curved) bulging in this case, with less than 2% error in cross-sectional area.

2.2 Kocks mills (Byon et al. 2017)

All three transition types are modeled as a circular arc with radius of curvature R_s and center offset P_c from the profile's center:

round → **Koval** $\frac{P_c}{B_0} = 3.133 \cdot \frac{F_H}{F_0}$, $R_s = B_1 - P_c$.

Koval → **Koval** $P_c = 0$, $R_s = B_1$ (a circle through the section's own center).

Koval → **round** P_c and R_s solve the system

$$R_s = \frac{DS}{2} + P_c, \quad (B_1 \sin 60^\circ)^2 + (B_1 \cos 60^\circ + P_c)^2 = R_s^2,$$

where DS is the current pass's own inscribed circle diameter (`roll_pass.inscribed_circle_diameter`) giving the closed form

$$P_c = \frac{(DS/2)^2 - B_1^2}{B_1 - DS}.$$

2.3 Implementation

The dispatch between these cases (module `pyroll.profile_bulging.three_roll_pass`) is driven purely by classifiers already available on the roll pass and its incoming profile – no additional configuration is required:

Condition	Flat roll	Curved (Koval) roll
"round" in <code>in_profile.classifiers</code>	Min circular arc	Byon round → Koval
"round" in <code>roll_pass.classifiers</code>	(not treated by Min)	Byon Koval → round
otherwise	Min linear chord	Byon Koval → Koval

The resulting eccentricity and radius of curvature are exposed as the hooks `ThreeRollPass.OutProfile.b` and `ThreeRollPass.OutProfile.bulge_radius` (the latter is `None` for Min's linear-chord passes, which have no single radius of curvature).

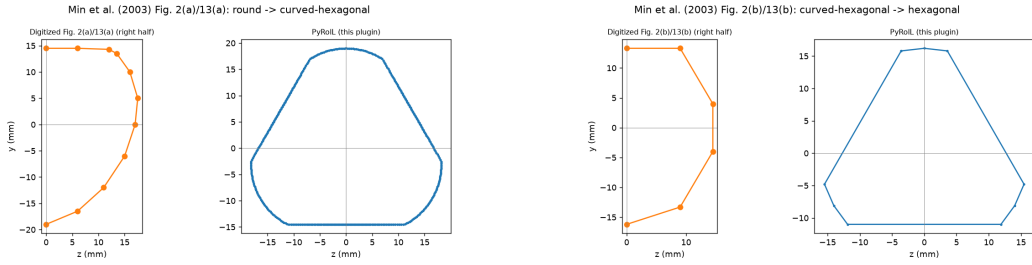
For the circular-arc cases, the free surface is traced directly, angle by angle, as whichever is closer to the profile's center: the nearest corner's bulge circle, or the roll's own unbulged contour (full roll contact). This is deliberately *not* built via the union of the three, generally overlapping, corner circles intersected with the roll contour, since that is prone to a spurious reentrant cusp wherever two such shapes cross without being exactly tangent there. Sampling the two candidate radii directly instead guarantees a shape with no self-crossings and keeps the residual non-smoothness to a minimum, but cannot remove it entirely: two neighboring corners' bulge circles are not, in general, tangent to each other or to the roll contour at the point where the model hands off between them, so a small change in slope at that point is an inherent feature of connecting three independently-fitted per-corner arc formulas into one closed cross-section - most visible for a heavily-reduced first (round-input) pass, where the eccentricity is large relative to the spacing between corners (see the round → Koval case in Fig. 2). The source papers validate each corner's own arc against a one sixth-symmetric sector of FE/experimental data (Section 4) and do not themselves address the smoothness of the reassembled full cross-section.

3 Two-roll passes

For completeness, and unchanged in this restructuring, the plugin also implements the round-oval-round free-surface model of Lee and Choi [3], Lee and Goldhahn [4], and Lee [5] and a circular-arc construction for square-diamond-square / oval-square passes (module `pyroll.profile_bulging.two_roll_pass`), dispatched analogously by the classifiers of the incoming profile and the current pass.

4 Validation

The three-roll models are validated against the figures of the source papers by reproducing their process parameters in PyRoll and comparing the predicted cross-section against boundary coordinates digitized from the printed figures. The out-profile width is pinned to the digitized shape's own spread tip radius for each case (bypassing spread prediction, see Section 1), so that only the free-surface model itself is being validated. See `tests/test_three_roll_min_flat.py`, `tests/test_three_roll_byon_koval.py` and `tests/test_three_` for the full test code and digitized data (`tests/data/`); running the test suite regenerates the comparison plots under `tests/output/`.



(a) Min et al., round $\rightarrow$ curved-hexagonal (circular arc). (b) Min et al., curved-hexagonal $\rightarrow$ hexagonal (linear chord).

Figure 1: PyRoll prediction (right) vs. digitized Fig. 2/13 of Min, Kwon, Lee, Woo, and Im [1] (left).

Coordinates were read off 300 dpi page renders of the source PDFs by eye against the printed axis gridlines; typical reading uncertainty is ± 0.5 –1 mm for the millimeter-scale shape plots and ± 0.01 –0.03 for the dimensionless spread/eccentricity calibration plots (Figs. 7, 9, 12). Byon's Figs. 8–10 plot one sixth-symmetric sector of the shape in the paper's own local axis orientation, which is not necessarily aligned with this plugin's own corner-angle convention; those comparisons are therefore shown side by side rather than overlaid. `tests/test_three_roll_bulge_formulas.py` additionally validates the eccentricity coefficients (2.40, 3.133) directly against the digitized FE calibration data of Figs. 7 and 12, and the closed-form Koval $\rightarrow$ round eccentricity against its defining equations, independent of any PyRoll geometry construction.

`tests/test_three_roll_end_to_end_spreading.py` complements this with a fully predictive (not pinned-width) demonstration: two-pass sequences for both mill types,

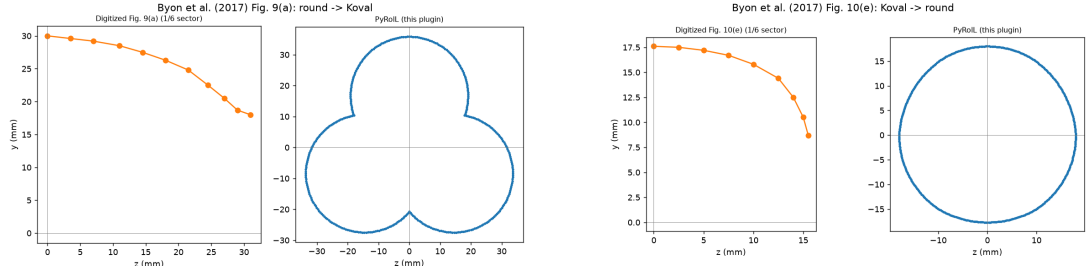
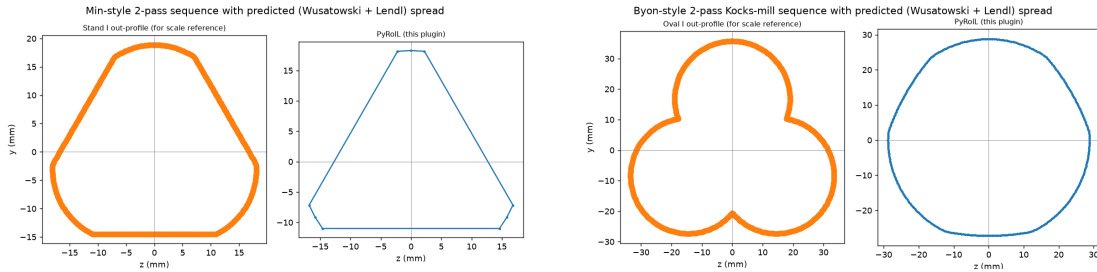
(a) Byon et al., round $\rightarrow$ Koval (pass 1).(b) Byon et al., Koval $\rightarrow$ round (pass 7, final).

Figure 2: PyRoll prediction (right) vs. digitized Fig. 9/10 of Byon, Kim, Kim, and Lee [2] (left), 7-pass Kocks mill.

with the out-profile width predicted by `pyroll-wusatowski-spreading` fed by `pyroll-lendl-equivalent-method` (Section 1), and this plugin's bulging model applying on top with no manual intervention, as a real user of the plugin would run it.



(a) Min-style flat-roll sequence.

(b) Byon-style Kocks-mill sequence.

Figure 3: Second-pass out-profile (right) with the width predicted end to end by Lendl + Wusatowski, shown against the first-pass out-profile for scale (left, not a reference shape here).

5 Usage instructions

Importing the plugin registers its post-processors on `TwoRollPass` and `ThreeRollPass`; no further setup is needed. Load a spread-prediction plugin and the `pyroll-lendl-equivalent-method` package alongside it (see Section 1) to have the out-profile width predicted rather than requiring it explicitly:

```
from pyroll.core import Profile, Roll, ThreeRollPass, FlatGroove
import pyroll.wusatowski_spreading
import pyroll.lendl_equivalent_method
import pyroll.profile_bulging
```

```
in_profile = Profile.round(diameter=35.8e-3, temperature=1323.15, strain=0,
```

```
material=["C45", "steel"], flow_stress=100e6,  
length=1, density=7.5e3,  
specific_heat_capacity=690, thermal_conductivity=23)  
  
roll_pass = ThreeRollPass(  
    roll=Roll(groove=FlatGroove(r1=5e-3, usable_width=45e-3, pad_angle=30),  
              nominal_radius=185e-3, rotational_frequency=248 / 60),  
    inscribed_circle_diameter=29e-3,  
)  
  
out_profile = roll_pass.solve(in_profile)  
print(out_profile.width, out_profile.bulge_radius, out_profile.cross_section.area)
```

Shell command to build this documentation (requires a working LaTeX installation):

```
pyroll input.py solve report
```

References

- [1] J. H. Min et al. “Analytical model for prediction of deformed shape in three-roll rolling process”. In: *Journal of Materials Processing Technology* 140 (2003), pp. 471–477. DOI: 10.1016/S0924-0136(03)00717-9.
- [2] Sang-Min Byon et al. “An approximate model to predict the surface profile of material sections in a 3-roll rolling process”. In: *Journal of Mechanical Science and Technology* 31.7 (2017), pp. 3489–3497. DOI: 10.1007/s12206-017-0638-7.
- [3] Youngseog Lee and Sangwoo Choi. “New Approach for the Prediction of Stress Free Surface Profile of a Workpiece in Rod Rolling”. In: *ISIJ International* 40.6 (2000), pp. 624–626.
- [4] Youngseog Lee and Gert Goldhahn. “Prediction Model of the Exit Cross Sectional Shape in Round-Oval-Round Pass Rolling”. In: *International Journal of the Korean Society of Precision Engineering* 2.1 (2001), pp. 86–94.
- [5] Youngseog Lee. “Prediction of the Surface Profile and Area of the Exit Cross Section of Workpiece in Round–Oval–Round Pass Sequence”. In: *ISIJ International* 42.7 (2002), pp. 726–735.